HANDOFF-FEASIBILITY AUDITS FOR HIERARCHICAL SKILL-CHAIN PLANNERS

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Paper under double-blind review

ABSTRACT

Hierarchical robot planners compose learned skills, options, and subgoals across long horizons. Their failure modes are often concentrated at handoffs: an option may terminate outside the next option's initiation support, accumulate transition drift, or produce an abstract state that looks valuable but is no longer executable. We study this as a handoff-feasibility audit for proxy-tail planning. A high-level proxy selects the most promising plan from a candidate pool, while public boundary diagnostics measure initiation violation, termination drift, reachability gap, boundary surprise, and estimated chain executability. In a controlled option-chain simulator, increasing candidate budget raises the selected proxy score by 7.52, but the selected chain's executability at $N = 512$ falls to 15.2% of its $N = 1$ value. A Handoff-Calibrated Sieve, using only public boundary evidence, improves large-budget true utility by 0.53 and improves executability. Oracle, calibrated, random, and anti-correlated controls show that the degradation is tied to proxy-boundary miscalibration rather than candidate count alone. The paper is a mechanism study and diagnostic target for real skill-chain planners, not a benchmark robotics claim.

1 INTRODUCTION

Long-horizon robotic behavior is often built by composing temporally extended skills. Temporal abstraction is useful because a planner can reason over a shorter sequence of options, but every option boundary introduces a fragile interface. The next skill may not be initiable from the previous skill's stochastic termination state; a large abstract jump may exceed the reliable span of the learned controller; and a high-level proxy may reward ambition without measuring handoff feasibility.

This paper studies the failure as a *handoff-feasibility audit*. We sample candidate option chains, score them with a high-level proxy, and inspect the selected proxy tail with public diagnostics. The audit asks whether the chosen chain remains executable as candidate budget increases, or whether selection concentrates on rare high-proxy plans whose boundary evidence is unsafe. The core contribution is not a generic reward-model overoptimization result. It is the option-specific mechanism: initiation support, termination drift, reachability, and boundary surprise jointly determine whether a selected skill chain can actually execute.

We make three contributions. First, we define a controlled option-chain simulator where boundary feasibility is observable and can be attacked directly. Second, we evaluate proxy-tail planning, handoff-calibrated selection, and multiple scorer controls across candidate budgets up to $N = 512$. Third, we include a rank-tail calibration check that predicts selected utility from a finite proposal population; it is used only as a diagnostic lens, not as the main novelty.

2 RELATED WORK

The options framework formalizes temporal abstraction through initiation sets, intra-option policies, and termination conditions (Sutton et al., 1999; Precup, 2000). Hierarchical RL methods such as MAXQ (Dietterich, 2000), surveys of HRL (Barto & Mahadevan, 2003), and option-critic (Bacon et al., 2017) study how to learn or compose such abstractions. Robot skill chaining and task-and-motion planning also emphasize preconditions, postconditions, and execution feasibility.

Reward-model overoptimization (Gao et al., 2023) explains why selecting strongly against an imperfect proxy can fail. This paper borrows that selection-pressure lens but does not repackage it as a new general theorem. The new question is where the proxy error comes from in hierarchical skill planning. We isolate the handoff boundary: initiation support, stochastic termination, reachability, and compounding abstract-state mismatch.

3 Option-Handoff Model

We use a controlled option world rather than a benchmark suite so the mechanism is observable. Each option o has an initiation center c_o, radius r_o, stochastic termination mean m_o, termination noise σ_o, nominal reward R_o, and reliable span d_o. Given an abstract state estimate x_t, initiation probability decreases with the margin $r_o - |x_t - c_o|$. Long options can appear attractive under nominal reward but become unreliable when their span exceeds d_o. Termination drift then propagates into the next handoff.

A plan $\tau = (o_1, \ldots, o_H)$ has hidden true executability $E(\tau)$, true utility $U(\tau)$, proxy score $S(\tau)$, and public diagnostics

$$D(\tau) = \{\text{initiation violation, termination drift, reachability gap, boundary surprise}\}.$$

The miscalibrated proxy rewards nominal value and ambitious subgoal distance while under-penalizing unsafe boundary evidence. This creates a selected-tail failure surface: the proxy can assign high score to chains whose handoffs are least likely to execute.

4 Audit And Calibration

The proxy-tail selector chooses the candidate with highest proxy score from a candidate pool. The handoff-calibrated selector ranks the same candidates by

$$S_{\text{handoff}}(\tau) = S(\tau) - \lambda B(D(\tau)),$$

where B combines initiation violation, termination drift, reachability gap, boundary surprise, and estimated chain executability. It uses no hidden execution labels, no true utility, and no oracle feasibility labels. Its purpose is diagnostic: if it repairs the failure, boundary evidence is the relevant axis.

Rank-tail calibration check. For a finite proposal population $P = \{(S_i, U_i)\}_{i=1}^{M}$ sorted by proxy rank, with deterministic tie-breaking, the selected true utility under independent with-replacement sampling satisfies

$$\mathbb{E}[U_{\text{selected}}(N)] = \sum_{r=1}^{M} U_{(r)} \left[\left(\frac{r}{M}\right)^N - \left(\frac{r-1}{M}\right)^N \right].$$

The identity simply states how selected mass concentrates on high proxy ranks. It does not prove degradation; degradation occurs only when the high-proxy tail has poor true utility. The experiment tests that empirical condition directly.

5 Experiments

We run four checks. First, proxy-tail planning is swept over $N \in \{1, 2, 4, 8, 16, 32, 64, 128, 256, 512\}$. Second, Handoff-Calibrated Sieve is evaluated on the same candidate sets. Third, controls replace the miscalibrated proxy with calibrated, oracle, random, and anti-correlated scorers. Fourth, the rank-tail calibration identity is compared with Monte Carlo selection from a finite proposal population. All claims are audited by `python -m skill_handoff_audit.experiments.run_claim_audit`.

Main result. At $N = 512$, proxy-tail planning has a much higher proxy score than at $N = 1$, but selected executability is only 15.2% of the $N = 1$ value. True utility peaks at small N and then falls below zero under the miscalibrated proxy. This supports the handoff-audit claim: candidate budget can amplify option-boundary errors rather than merely improving abstract plans.

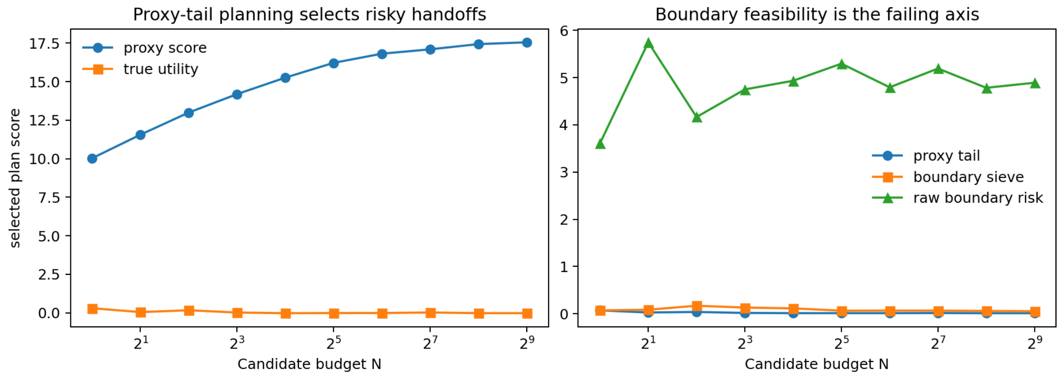

Figure 1: Proxy-tail planning increases selected proxy score while moving toward less executable option chains. Boundary risk exposes the failing axis.

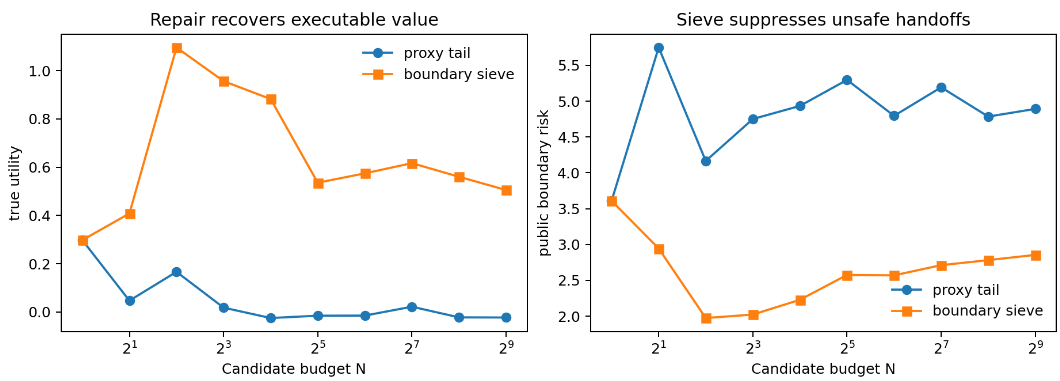

Figure 2: Handoff-Calibrated Sieve selects lower-risk chains from the same candidate sets and recovers large-budget true utility.

Repair. Using the same candidates, Handoff-Calibrated Sieve improves large-budget true utility by 0.53, improves executability, and reduces public boundary risk. The repair is deliberately simple; it demonstrates that public handoff evidence is the right diagnostic axis in this controlled setting.

Controls. The oracle feasibility scorer does not show the same failure and improves with N. The calibrated scorer is substantially more stable than the miscalibrated scorer. Random and anti-correlated scorers behave as negative controls. These controls suggest that the failure is not caused by candidate-set size alone.

6 Limitations

The study is controlled and low-dimensional. The option diagnostics are hand-designed, and the sieve is not claimed to be a final robot algorithm. The results should not be read as benchmark evidence that one hierarchical planner outperforms another. They show a mechanism that real systems should measure: whether high-level proxy-tail planning is selecting plans from the unsafe handoff boundary.

7 Conclusion

Proxy-tail planning can be unsafe for hierarchical skill planners when high-level scores are miscalibrated at option handoffs. The simulator exposes the mechanism, the rank-tail calibration check explains the selection pressure, and Handoff-Calibrated Sieve demonstrates that public boundary di-

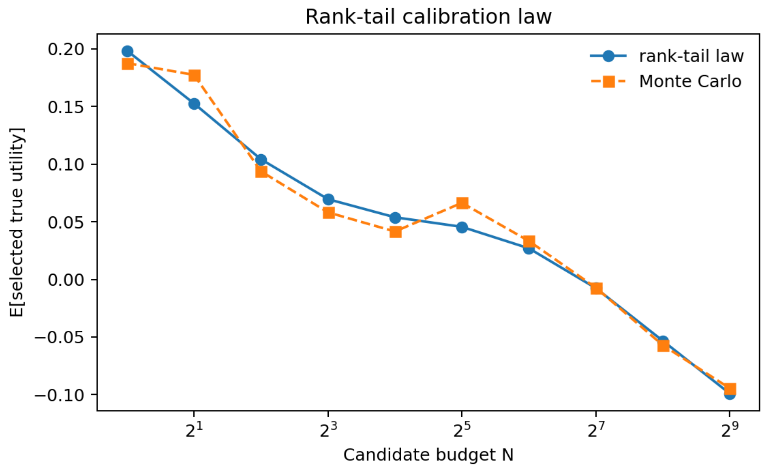

Figure 3: The rank-tail calibration identity matches Monte Carlo selection. Current mean absolute error is 0.0106.

agnostics can repair much of the large-budget degradation. The next step is to learn these diagnostics from robot skill data and evaluate them on benchmark long-horizon manipulation tasks.

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A Reproducibility

The public commands are:

```
python -m skill_handoff_audit.experiments.run_smoke
python -m skill_handoff_audit.experiments.run_all
python -m skill_handoff_audit.experiments.run_claim_audit
pytest
```

The claim audit writes `docs/claims.md` and `results/claim_audit.json`.

B CLAIM BOUNDARY

The paper claims a handoff-boundary mechanism in a controlled simulator. It does not claim real-robot benchmark superiority, universal degradation for all hierarchical planners, or that the proposed sieve is optimal.